No doubt, when it comes to extra dimensions, I'm biased. I've worked on aspects of extra dimensions for more than fifteen years, so they hold a special place in my heart. But, with that confession as a qualifier, it's hard for me to imagine a discovery that would be more exciting than finding evidence for dimensions beyond the three with which we're all familiar. To my mind, there is currently no other serious proposal whose confirmation would so thoroughly shake the foundation of physics and so thoroughly establish that we must be willing to question basic, seemingly self-evident, elements of reality.

The Higgs, Supersymmetry, and String Theory

Beyond the scientific challenges of searching into the unknown, and the chance of finding evidence of extra dimensions, there are a couple of specific motivations for recent upgrades on the accelerator at Fermilab and for building the mammoth Large Hadron Collider. One is to find Higgs particles. As we discussed in Chapter 9, the elusive Higgs particles would be the smallest constituents of a Higgs field—a field, physicists hypothesize, that forms the Higgs ocean and thereby gives mass to the other fundamental particle species. Current theoretical and experimental studies suggest that the Higgs should have a mass in the range of a hundred to a thousand times the mass of the proton. If the lower end of this range turns out to be right, Fermilab stands a reasonably good chance of discovering a Higgs particle in the near future. And certainly, if Fermilab fails and if the estimated mass range is nonetheless correct, the Large Hadron Collider should produce Higgs particles galore by the end of the decade. The detection of Higgs particles would be a major milestone, as it would confirm the existence of a species of field that theoretical particle physicists and cosmologists have invoked for decades, without any supporting experimental evidence.

Another major goal of both Fermilab and the Large Hadron Collider is to detect evidence of supersymmetry. Recall from Chapter 12 that supersymmetry pairs particles whose spins differ by half a unit and is an idea that originally arose from studies of string theory in the early 1970s. If supersymmetry is relevant to the real world, then for every known